Course Description

Title of Course: Open Source Software Lab **Course Code: CS213**
L-T-P Scheme: 0-0-2 **Course Credit: 1**
Course Coordinator: Dr. Animesh Kumar Dubey

This Lab will be based on the subject run in CSE-Elective -1

Prerequisites

Students would have already studied the basic concepts of Python and machine learning (ML), as well as their applications.

Objectives

1. To introduce the basic concepts and techniques of Artificial Intelligence (AI) and the need of AI techniques in real-world problems
2. To provide knowledge of various Machine and Deep learning algorithms and the way to evaluate performance of these algorithms.
3. To explain the concept of different Auto-encoders.

Learning Outcomes:

CS213: Open Source Software Lab	
Course Outcome	**Description**
CO1	Students will demonstrate knowledge of basic ML and DL models and the need of AI techniques in real world problem.
CO2	To evaluate the CNN models on different parameters for image data classification.
CO3	To implement RNN and LSTM to learn, predict and classify the real-world problems in the paradigms of deep learning.
CO4	To learn the concept of Auto-encoders and enhancing Generative Adversarial Networks (GANs) using auto-encoders.

Course Content

Lab Exercise (1-4)	Study and practical implementation based on various ML algorithms for solving real world problems
Lab Exercise (5-8)	Study and practical implementation based on DL algorithms for some large data sets.
Lab Exercise (9-10)	Study the concept, working and practical importance of Auto-encoders.

Text books

1. Hands-On Machine Learning With Scikit-Learn, Keras, and Tensorflow, *Aurélien Géron,* 2019
2. Fundamentals of Deep Learning, *Nithin Buduma, Nikhil Buduma, and Joe Papa*, 2017.

References

1. Deep Learning: A Practitioner's Approach, *Josh Patterson and Adam Gibson*, 2017.
2. Neural Networks and Deep Learning, *Charu C. Aggarwal*, Springer, 2018.

Online course link:

https://onlinecourses.nptel.ac.in/e-learning/preview/noc26_ee189 *(Machine Learning and Deep learning - Fundamentals and Applications, by Prof. M. K. Bhuyan, IIT Guwahati, starting from 20th of July 2026)*

https://onlinecourses.nptel.ac.in/e-learning/preview/noc26_cs181 *(Deep Learning for Natural Language Processing, by Prof. Pawan Goyal, IIT Kharagpur, starting from 20th of July 2026)*

Evaluation Scheme

Evaluations	**Marks**
P1	15 Marks
P2	15 Marks
Lab Work	40 Marks
Lab Record	15 Marks
Attendance and Discipline in Lab	15 Marks
Total	**100 Marks**